Training-free Enhancement of SDXL

FreeU · PAG · DUNE: what helps, and at what cost

[Name A] · [Name B] · [Name C]

Generative AI Technical Review · November 2026



SDXL + PAG, prompt p130

SDXL still draws broken structure



Two giraffes with fused bodies and extra legs



“A man holding a bottle”: three bottles, tangled hands

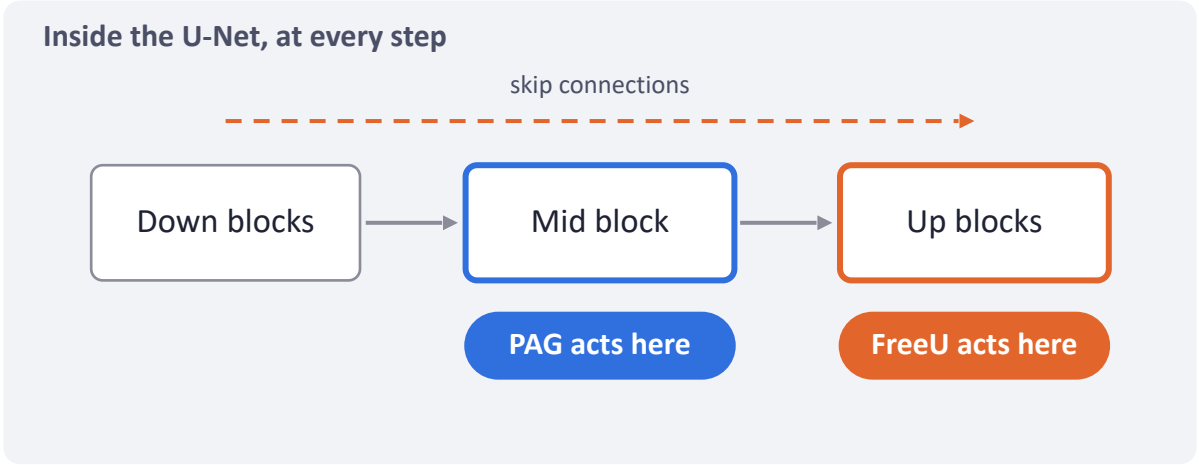
54%

of base-SDXL judgements were tagged with a structure or limb error

Retraining a 3.5B-parameter model is out of reach. Can we fix this at inference time instead?

Where the two fixes plug in





FreeU: rebalance structure and detail

The U-Net decoder mixes two signals:

- Backbone features carry layout and meaning
- Skip connections carry fine, high-frequency detail

When detail dominates, structure can fall apart. FreeU turns the backbone up and the skip detail down.

SDXL settings we used

Backbone ×1.3 / ×1.4



Skip detail ×0.9 / ×0.2



+3%

inference time: four multiplications per block

PAG: learn from a broken prediction

1

Predict as usual

The model predicts the noise to remove, guided by the text.

2

Predict again, broken

Same step, but self-attention is replaced by the identity: the layout falls apart.

3

Move away from it

Push the sample away from the broken prediction (scale 3).

+47%

inference time: one extra U-Net pass per step

Like showing the model a deliberately bad draft and saying “not like this”.

Same model, prompts and seeds

150
COCO captions, 5 categories

5
settings incl. our DUNE reimplementation

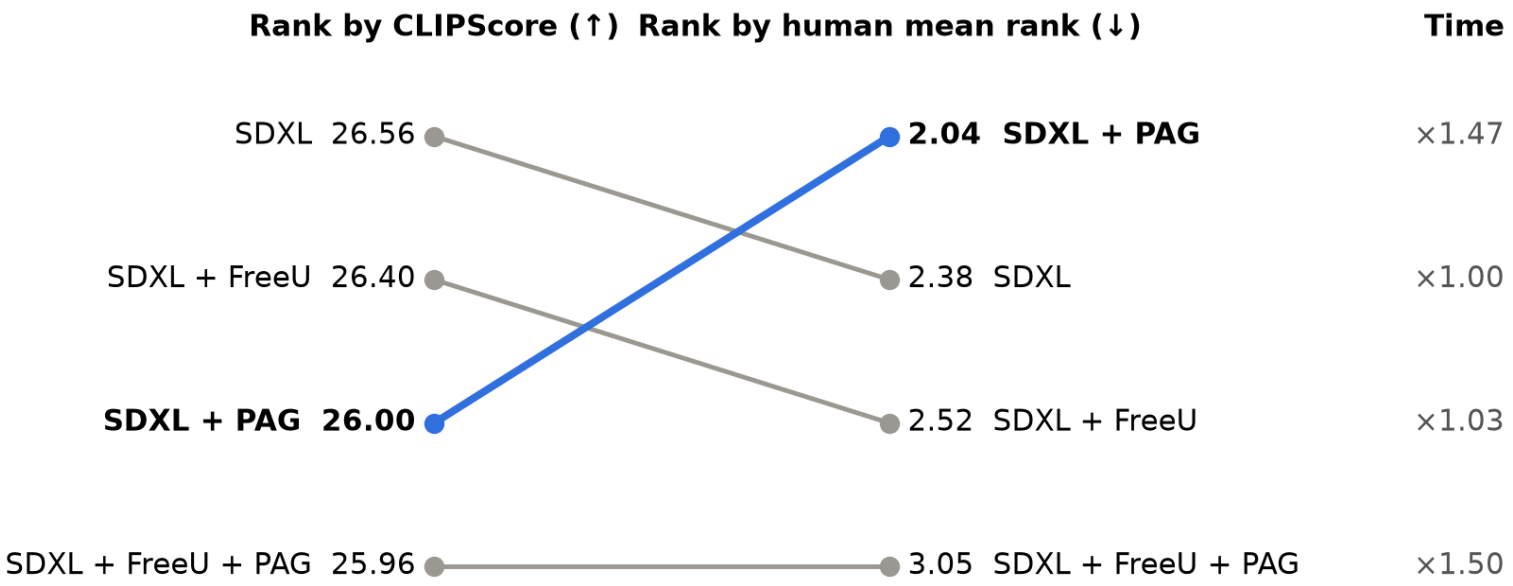
1,000
images on one NVIDIA A40

3 × 60
blind four-way human rankings

How we measured

- CLIPScore: does the image match the text?
- Similarity to the plain SDXL image: how much did the method change it?
- Time and memory per image
- Humans: pick best and worst of 4 shuffled images, tag failures

The metric and the raters disagree



CLIPScore only checks whether the named objects are present; it cannot see a fifth leg.

PAG fixes structure; stacking saturates

	Structure / limb error	Over-saturation / artifact	Mismatch with description	Missing object	Duplicated object
SDXL	**54%**	7%	14%	9%	7%
SDXL + FreeU	30%	50%	15%	7%	8%
SDXL + PAG	26%	12%	14%	6%	9%
SDXL + FreeU + PAG	27%	**82%**	12%	6%	5%

Share of judgements tagged with each failure

54→26%

structure errors, SDXL → PAG

7→82%

over-saturation, SDXL → FreeU + PAG

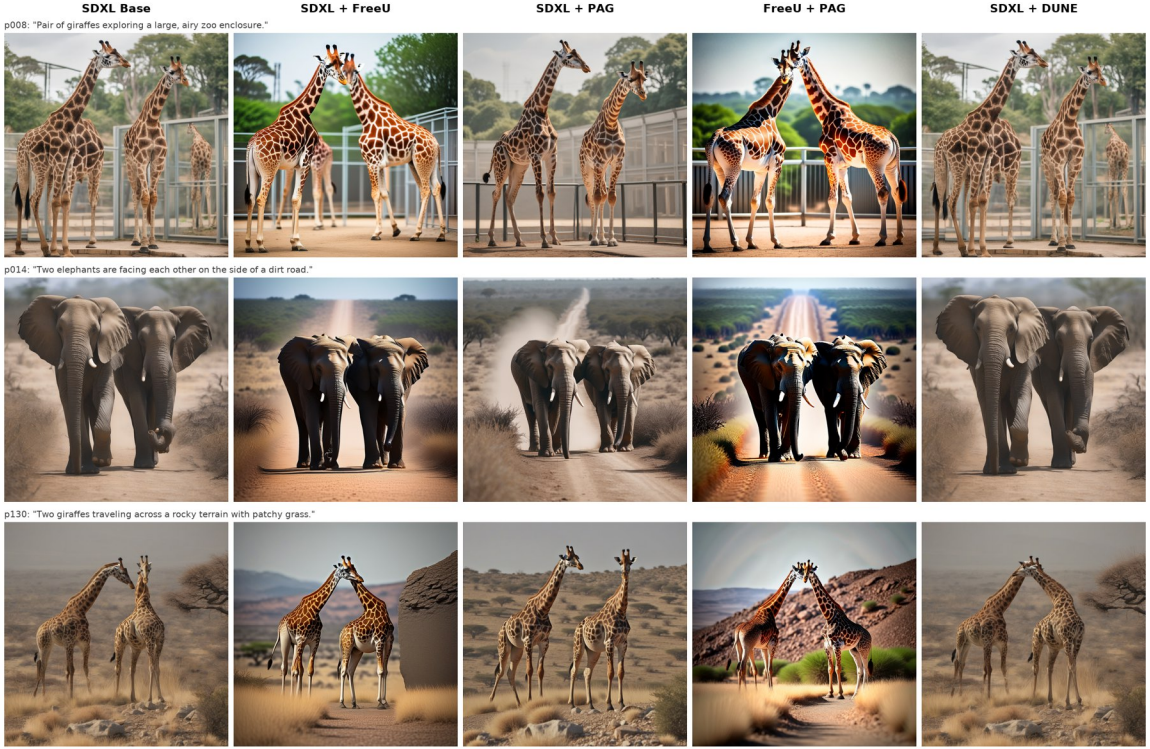
5%

of judgements picked PAG as worst (SDXL 22%)

42%

picked FreeU + PAG as worst

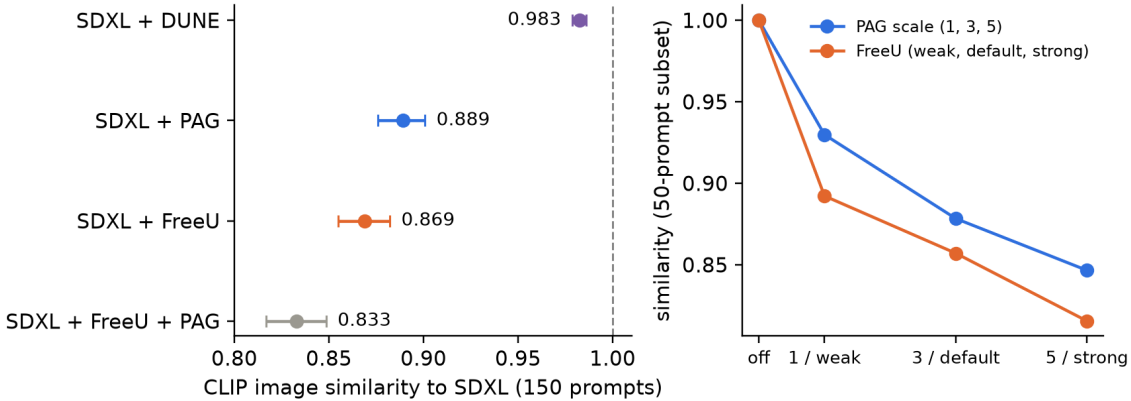
Two animals: where PAG helps most



Columns: SDXL, + FreeU, + PAG, + FreeU + PAG, + DUNE

- SDXL and FreeU fuse the two giraffes and add legs
- PAG separates them cleanly with natural colour
- FreeU + PAG: right shapes, poster-like colour

DUNE (2026): the most conservative fix





0.983
similarity to SDXL

×1.00
inference time

−0.04
CLIPScore, not significant

What limits our conclusions

Few raters

Three raters, all authors, with low agreement (Fleiss' $\kappa = 0.105$). One preferred plain SDXL, another PAG.

Metric blind spot

CLIPScore misses structure, so the automatic results understate PAG.

Narrow scope

150 short COCO captions, one sampler, 30 steps, CFG 5.

DUNE

Our own reimplementation of an unreviewed preprint, rated by metrics only.

Takeaways

1. Use PAG when structure matters (≈1.5× time)
2. Do not stack FreeU and PAG at default strengths
3. Do not judge these methods by CLIPScore alone



Code, prompts, ratings: github.com/AlMuM/sdxl-freeu-pag